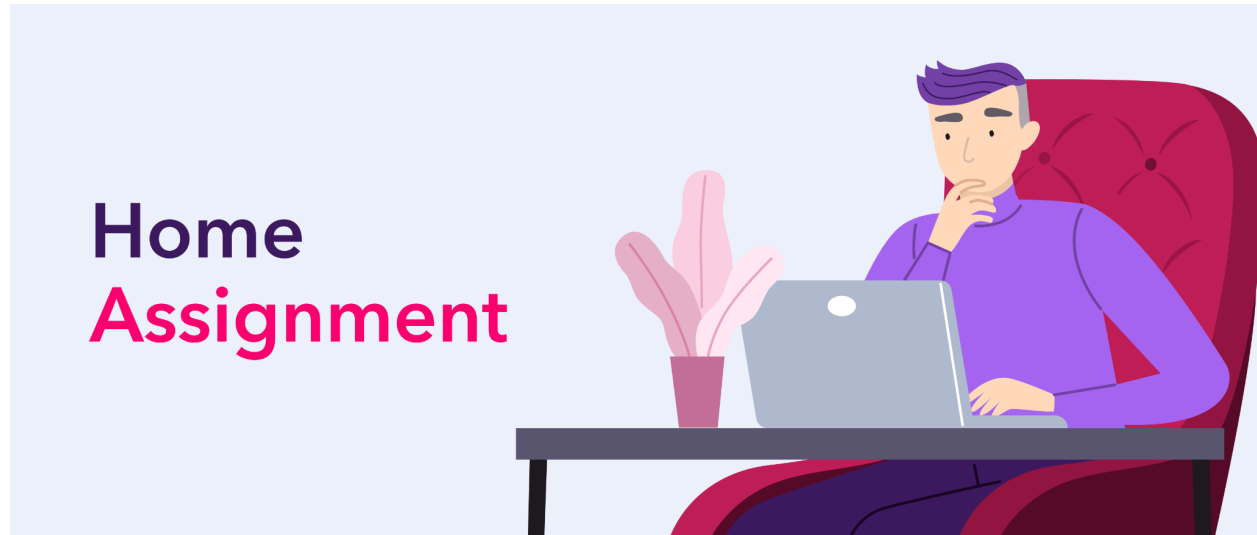




AI Engineer - Home Assignment

Build an MCP server that lets a user ask natural-language questions about a dataset and get accurate answers. We've chosen the dataset options and the deployment target (local) so you can focus on the solution itself and business case.

Data:

Pick one dataset (both are on Hugging Face).

A. Customer Support Tickets — [Support Dataset](#)

B. Customer Churn — [Churn Dataset](#)

Pick whichever problem you find more interesting or relevant to your past experience.

Specs:

MCP Server: runnable locally with Claude code or Codex

Tools and capabilities = your call

Language / stack = your call

Scope = your call (RAG/SQL/Other)

Model/API = Any LLM is fine. If you use an API, read it with an environment variable (ex. ANTHROPIC_API_KEY or OPENAI_API_KEY). Document in your README which model you used and how to configure it.

Pick the solution that fits the dataset and defend it.

**You are encouraged to use AI tools to create this solution!*

Deliverables:

1. GitHub Repository

- a. Code that can be run locally
- b. README that explains how to run it. Required variables, packages, how to run and how to connect it to Claude Code / Codex (MCP configuration)
- c. Repository should be publicly available

2. One-Page PDF Design Document

- a. Design: what you built and how it works to answer questions
- b. Why: why you chose this solution, and what you deliberately chose to avoid
- c. Production: What would you change if this were a real production environment
- d. Risks: what could go wrong, how would you prevent it
- e. Business Impact: How would you create value with this solution at CHEQ
- f. Keep the document concise to one page

Scoring:

The build must actually run and answer simple questions correctly. We are mainly focused on the logic of your solution (in the One-Page document), creativity and business sense.

Submission:

Send an email with the link to your repo and PDF to zack.s@cheq.ai, robert.t@cheq.ai and avishag.y@cheq.ai.

Timeline:

Reply to this email with how long this task will take you and when you'll deliver. Feel free to ask zack.s@cheq.ai if anything is unclear.

Good luck!